CÉCILE LE DIZES

PhD in Geophysical Fluid Dynamics

cecile.ledizes@gmail.com
https://github.com/cecileledizes
+33 6 28 07 51 08
Toulouse, France

I am a researcher in geophysical fluid dynamics and physical oceanography, with expertise in numerical modelling and scientific computing. I am interested in understanding how small-scale ocean processes control energy exchanges and mixing, and in developing physically based parameterizations for ocean and climate models.

EDUCATION

2022 - 2025 **Ph.D. | Université de Toulouse, France**
Three-dimensional modeling of internal waves generation by isolated seamounts.

2018 - 2022 **Engineering Diploma (Master's eq) | ISAE-SUPAERO, Toulouse, France**
Major in fluid dynamics : turbulence, compressible flows, aerodynamics ; Minor in applied mathematics : numerical simulations, data assimilation ; Elective courses in climate physics.

2021 - 2022 **Master in Fluid dynamics, Energy and Transfers (DET) | Université de Toulouse, France**
Classes on hydrostatic instabilities, aerosols and suspensions, porous media, aeroacoustic...

2016 - 2018 **Preparatory Classes to Engineering Schools | Lycée Thiers, Marseille, France**
Theoretical formation in mathematics, physics and chemistry in order to prepare the national exam test to enter the French Engineering Schools.

RESEARCH EXPERIENCE

Oct 2022 - Nov 2025 **Ph.D. | Institut de Mécanique des Fluides de Toulouse (IMFT), Toulouse, France**
Three-dimensional modeling of internal waves generation by isolated seamounts - with Matthieu Mercier, Nicolas Grisouard and Olivier Thual.

- Development of a 3D Boundary Element Method (BEM) to solve the problem of internal tide generation by isolated seamounts, in both the subcritical and supercritical regimes. Comparison with high-resolution GPU simulations using the Julia software *Oceananigans*.
- Co-supervision of a master student for the application of the BEM to realistic bathymetry and tides.
- Collaboration with N. Grisouard during a two-months visit at the University of Toronto.
- Funding: competitive national doctoral fellowship (MESRI) ; TESS Booster and TESS mobility grants.
- Elected Ph.D. students representative at the doctoral school of Sciences de l'Univers, de l'Environnement et de l'Espace (SDU2E) ; and at the laboratory council.

June 2024 - Aug 2024 **GFD fellow | Woods Hole Oceanographic Institution (WHOI), USA**
Density staircases in stratified rotating turbulence - with Jim McElwaine, Claudia Cenedese and Pascal Garaud.
Ten-week summer program with lectures on Multiscales Geophysical Fluid Dynamics and an independent research project investigating the formation and evolution of density layers in a stratified rotating turbulence, using both laboratory experiments and direct numerical simulations with the Fortran code PADDI.

April 2022 - Sept 2022 **Master's thesis | DAMTP, Cambridge University, UK**
Flushing of a dense cavity by a gravity current - with Stuart Dalziel.
I designed a laboratory experiment and ran 2D OpenFoam simulations to study entrainment and transport by a gravity current over a dense cavity.

Feb 2020 - Aug 2020 **Intern | Airbus Helicopters, Marignane, France**
Research project, development of a simple model for the icing of helicopter blades.

Sept 2019 - Jan 2020 **Intern | Institut de Recherche en Astrophysique et Planétologie (IRAP), Toulouse, France**
Data analysis of the oscillations of the delta-scuti star Altaïr - with Michel Rieutord.

TEACHING EXPERIENCE

INP-ENSEEIHT (2022-2025): Graduate teaching assistant for the course 'Introduction to turbulence' at first-year Master's level (96h) and the course 'Linear elasticity' at third-year undergraduate level (32h). Supervision of student projects involving experimental (PIV, hot-wire anemometry) and numerical (RANS) methods.

ISAE-SUPAERO (2023-2024) : Lectures on internal waves for the course 'Geophysical flows' at M1 level (15h).

PUBLICATIONS AND OUTREACH

Le Dizes, C., Mercier, M. J., Thual, O., & Grisouard, N. (2026) Barotropic adjustment to seamounts under rotation and deviations of internal tides direction, *in preparation.*

Le Dizes, C., Grisouard, N., Thual, O., & Mercier, M. J. (2025). Three-dimensional modelling of internal tide generation over isolated seamounts in a rotating ocean. Journal of Fluid Mechanics, 1022, A5. doi:10.1017/jfm.2025.10647

Le Dizes, C., McElwaine J., Cenedese C., & Garaud P. (2024). Mixing and Layering in Stratified Rotating Turbulent Flow, WHOI proceedings, p.253. https://gfd.whoi.edu/wp-content/uploads/sites/18/2025/12/2024-GFD-Volume-Online-version-reduced.pdf

Le Dizes, C., Mercier, M., Grisouard, N., & Thual, O. (2024). Comprendre les ondes océaniques pour mieux prévoir l'évolution du climat. The Conversation. https://theconversation.com/comprendre-les-ondes-oceaniques-pour-mieux-prevoir-levolution-du-climat-240030

Le Dizes, C., Rieutord, M., & Charpinet, S. (2021). Seismology of Altair with MOST. Astronomy & Astrophysics, 653, A26. https://doi.org/10.1051/0004-6361/202141291

CONFERENCES, WORKSHOPS AND SUMMERSCHOOLS

GDR MFGA et Navier-Stokes 2.0, Toulouse, France, 2026: Poster presentation

GDR Mécanique des Fluides géo- et astro-physique, Marseille, France, 2025: Oral presentation, 'Three-dimensional modelling of internal tide generation'.

Workshop 'Parametrisation in Climatology, Astrophysics and Planetology', Les Houches, France, 2025: Oral presentation, 'Three-dimensional modelling of internal tide generation'.

Seminar at the University of Toronto: Oral presentation, 'Density layering in rotating stratified turbulence'.

GFD summer program 'Multi-scale GFD', June-August 2024, WHOI, USA: Two weeks lectures on Multiscale Geophysical Fluids Dynamics. Final oral presentation: 'Density layering in rotating stratified turbulence'.

EGU Meeting 2024, Vienna, Austria: Poster presentation, '3D modelling of internal tides generation'.

GdR Math-GeoPhy, Nov 2023, Amiens, France: Oral presentation, '3D modelling of internal tides generation: a boundary element method for any localized topographies'.

Symposium LEFE/MANU, June 2023, Météo-France, Toulouse, France: Poster presentation, '3D modelling of internal tides generation: a Green's function approach for any localized topographies.'

Workshop 'Singularities and attractors in rotating and stratified fluids', Marseille, France.

SKILLS

Numerical
- Programming languages: Fluent in Python and Julia ; Strong skills in Matlab and C; basic skills in Fortran and Java.
- Methods: boundary element methods, DNS, subgrid-scale closures, RANS
- High performance computing: parallel computing with MPI, OpenMP; GPU using CUDA.
- Codes and tools: Oceananigans, OpenFOAM, PADDI, Ansys, LaTeX, Excel, PowerPoint.

Experimental
- Fluids mechanics experiments with stratified and rotating setups.
- Particle Image Velocimetry, hot-wire anemometry, dye-based light attenuation.

Languages French (native speaker), English (fluent), spanish (basic skills).